Path Planning and Foothold Adaptation for Quadrupedal Locomotion on Challenging Terrain

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Abstract. In this work, we propose a combined path planning algorithm and foothold adaptation for quadrupedal locomotion in challenging terrain. The proposed path planning algorithm, based on Rapidly-exploring Random Tree* (RRT*), efficiently determines kinematically feasible paths for quadruped considering the given terrain information. However, following a body reference alone is insufficient to overcome the entire of challenging terrain, as the path planning could not capture whole scenarios, the robot must adapt its plan based on its current situation. Therefore, we use reactive foothold adaptation and generate collision-free swing leg trajectories aligned with the adapted foothold positions. Through the experiments in a 3D simulation environment featuring a sparse stepping stone map, we demonstrate the efficacy of our combined framework, comprising path planning and foothold adaptation.

Keywords: Path Planning · Foothold Adaptation · Model Predictive Control · Quadruped Locomotion

1 Introduction

Recently, legged robots have shown their dynamic capabilities based on optimal control frameworks, specifically Model Predictive Control (MPC) [1,4,9]. Extensive applications on various legged robot platforms have demonstrated the impressive performance of MPC, encompassing high-speed running [12], robust walking [3], and complex dynamic motions. However, most existing research primarily focuses on maintaining robot balance and robustly tracking commands based on dynamics, while disregarding the influence of the surrounding environment. To enhance generality and achieve diverse locomotion capabilities, there is a growing need to investigate terrain-aware locomotion strategies. This work is in line with this motivation and proposes a hierarchical framework that combines a global path planner that finds out kinematically feasible body reference trajectory with subsequent local foothold adaptation during motion execution for stable locomotion of quadruped.

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1.1 Related Work

Terrain Aware Locomotion, Combined Approach. For terrain-aware locomotion, it is crucial to consider the selection of appropriate foothold positions to navigate through different terrain conditions. However, addressing foothold position alone poses challenges in terms of tractability as it disregards the inherent coupling between foothold position and the dynamics of body motion [2]. The coupled property between body motion and foothold position arises from that the robot relies on ground reaction forces to control its body in a desired manner while maintaining stability in the selected foothold position. Hence, one approach is the coupled approach, wherein the model predictive control problem plans the body motion and foothold positions together [2,8,10]. In this approach, the traversable region is given through segmented planes, which are utilized as explicit constraints for determining the foothold positions. By incorporating these constraints, the optimization process jointly optimizes the body trajectory and foothold positions concurrently, taking into account the whole body dynamics. Due to the model complexity including whole body dynamics, kinematic limit, and constraints from the surrounding map, these frameworks are hardly reproduced in other research groups.

Terrain Aware Locomotion, Hierarchical Approach. Another commonly used approach in terrain-aware locomotion is the hierarchical approach, where the process is divided into two stages: foothold planning and the generation of dynamically feasible body motion [6,11,12,15]. This approach is widely adopted due to its simplicity, as researchers often incorporate foothold adaptation into an existing framework while reusing their previous framework for generating body motion. In this approach, the foothold position is reactively adapted based on the commanded body trajectory within the traversable terrain. Subsequently, a dynamically feasible body motion is generated by solving an optimization problem that considers only the dynamic equations under the predefined foothold positions. However, using the predefined foothold positions can introduce potential challenges in the dynamic perspective, such as a limited support polygon, because the foothold planning is conducted without considering the body's dynamics explicitly. Therefore, further work makes the desired body configuration through pose optimization to guarantee the stability under the given foothold position [7], or makes the desired body reference that maximizes the potential adapted foothold region to make the robot more robust under the uncertainty [6].

Global Planning. The above works primarily focus on local planning strategies for quadruped locomotion on challenging terrains, aiming to determine appropriate actions for maintaining balance and accurately tracking commanded velocities. To fully utilize the capabilities and extend the range of applicability of local controller performance, researchers have developed global planners that determine reference body position trajectories from the initial to the goal position,

considering specific feasibility criteria such as dynamic feasibility [13] or obstacle free [5]. [13] uses the RRT-connect method to find out dynamically feasible paths, which are subsequently utilized as reference trajectories for the model predictive control [14]. To reduce computational complexity, they employ simplified dynamics models and sample dynamically feasible new nodes by integrating sampled control inputs. In [5], the A* method is utilized to identify feasible paths and avoid obstacles, between the start and goal position, and these obtained paths are then employed as references for the low-level model predictive control (MPC) framework.

1.2 Proposed Approach

Building upon our existing model predictive control (MPC) framework, we integrate a global path planner and a local foothold adaptation module to form a hierarchical framework designed to navigate challenging terrains, such as sparse stepping stones with varying heights, using provided terrain information. While our previous MPC framework performed effectively on flat and sloped terrains without visual feedback, it faced limitations in more complex scenarios, especially during vision-based walking, where the absence of a high-level planner restricted its ability to manage multiple steps or complex obstacles.

In this work, we first employ the RRT* path planning algorithm to generate a kinematically feasible reference trajectory from the start to the goal position, ensuring that potential foothold positions align with body configurations. The robot then follows this trajectory with reactive foothold adaptation, maintaining valid foothold positions during the stance phases. This integrated approach enables the robot to traverse challenging terrains by dynamically adjusting foothold positions in response to the kinematically feasible planned trajectory.

2 Architecture Overview

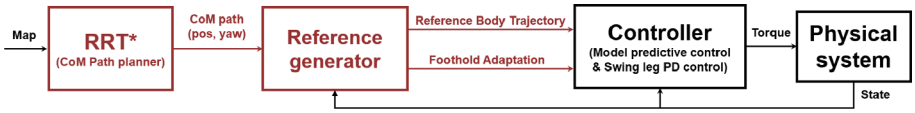

Fig. 1. Illustration of the whole architecture, the colored box indicates the components that are addressed in this study.

Given the starting point and terrain information, the RRT* algorithm computes a kinematically feasible path of body nodes for the robot. Each body node specifies the robot's body position and yaw, ensuring that valid foothold positions are available nearby. The solution path derived from the RRT* algorithm is then passed to a reference generator that operates concurrently with the controller.

This generator produces a reference body trajectory by interpolating between successive nodes along the path. The reference generator also derives reference foot trajectory; however, it does not directly use the footholds corresponding to body nodes in the path planning phase. Instead, it applies local adaptation to footholds derived from Raibert heuristics.

The proposed framework uses both the body reference and the adapted foothold positions to address an optimal control problem using a baseline controller [9]. This baseline controller resolves the nonlinear MPC problem to create a dynamically feasible motion that minimizes the MPC cost (regulating error of state and control input) for the given foothold positions. The ground reaction forces derived from the solution are then applied to the stance leg. Additionally, a swing leg PD controller guides the foot toward the next foothold position. A full overview of this framework is presented in Fig. 1.

3 Path Planning for Kinematic Feasible Path

The RRT* (Rapidly-exploring Random Tree) algorithm is a probabilistic roadmap-based motion planning algorithm widely used in robotics and autonomous systems (Algorithm 1). It efficiently constructs a tree-based representation of the configuration space to find an optimal path between a start and goal configuration. Unlike its predecessor, the RRT algorithm, RRT* aims to improve the optimality of the generated paths by iteratively rewiring the tree structure.

Algorithm 1: RRT* path planner pseudo code

```
Input  : x_init, φ
Output: Path(x_new, V)
V ← {(x_init, φ)}
for i = 1, 2, ..., n_max do
    x_rand ← SampleFree()
    x_nearest ← Nearest(x_rand, V)
    x_new ← Steer(x_nearest, x_rand)
    if ObstacleFree(x_new) = true then
        χ_near ← Near(x_nearest, V, n_neighbors)
        x_parent ← ChooseParent(x_new, χ_near, V)
        V ← V ∪ {(x_new, x_parent)}
        Rewire(χ_near, x_new)
        if Goal(x_new) = true then
            break
        end
    end
end
return Path(x_new, V)
```

The function *ObstacleFree* checks whether there are stones that can put all four feet in the given state x_{new} (Algorithm 2). If it exists, the state x_{new} is feasible, and if there is no stone to step on even one foot, the state x_{new} is judged to be colliding with an obstacle in RRT* algorithm (infeasible) as shown in Fig. 2. The position to put each foot, obtained here, is not entered into MPC as a foothold reference.

Algorithm 2: ObstacleFree for kinematic feasibility

```
Function ObstacleFree(x_new):
    for i = 1,2,3,4 do
        Foot_i ← Foot(i, x_new)
    end
    if CheckNearestStone(Foot_i) = true for all i then
        return true
    else
        return false
    end
return
```

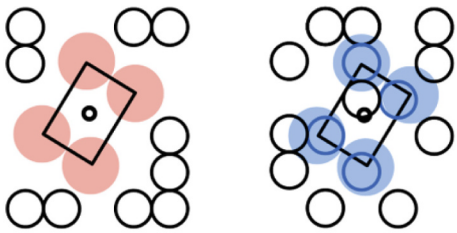

Fig. 2. Left : Kinematically infeasible case, Right : Kinematically feasible case

Since our RRT* path planner only considers kinematic feasibility for computational efficiency without considering dynamic feasibility, it is important to obtain a path as smooth as possible without sharp curvature. Therefore, we can obtain a sufficiently smooth path by setting the length of the robot's CoM path and yaw difference as the *cost* and optimizing nearly through the RRT* algorithm (Algorithm 3, 4).

While the simple *ObstacleFree* function presented in this paper ensures kinematic feasibility, it does not guarantee dynamic feasibility. For scenarios involving locomotion on relatively static and less challenging terrains, path planning without considering dynamic feasibility is feasible. However, when dealing with terrains that necessitate dynamic feasibility considerations (e.g. overcoming large gap), the *ObstacleFree* function presented in this paper is unable to generate a feasible path.

Algorithm 3: ChooseParent function

```
Function ChooseParent(x_new, χ_near, V):
    x_parent ← {φ}
    cost_min ← ∞
    for x_near ∈ χ_near do
        cost ← Cost(x_near) + distance(x_near, x_new)
        if cost < cost_min then
            cost_min ← cost
            x_parent ← x_near
        end
    end
    return x_parent
return
```

Algorithm 4: Rewire function

```
Function Rewire(χ_near, x_new):
    for x_near ∈ χ_near do
        cost_new ← Cost(x_new) + distance(x_new, x_near)
        if cost_new < Cost(x_near) then
            x_parent ← Parent(x_near)
            Replace the parent of x_near with x_new in V
        end
    end
return
```

4 Foothold Adaptation

This section explains a foothold adaptation method for the quadruped robot to follow the node given in path planning. Due to the computational load of the path planning process, it is challenging to plan all footholds during locomotion, and the robot may not be able to follow the planned body nodes and footholds perfectly. Fixed foothold positions without adjustment would fail to meet the robot's evolving motion requirements effectively.

4.1 Foothold position

We used modified Raibert's heuristic to obtain a nominal foothold position. Raibert's heuristic is widely utilized because it ensures that each foot lands beneath the corresponding hip during the middle of the stance phase, assuming the robot is moving at a constant velocity. The robot foot positioned below the hip, on average over time during the stance phase, helps to stabilize the body by allowing each foot of the robot to do a balanced stance. The next foothold position is first determined by the original Raibert's heuristic as follows.

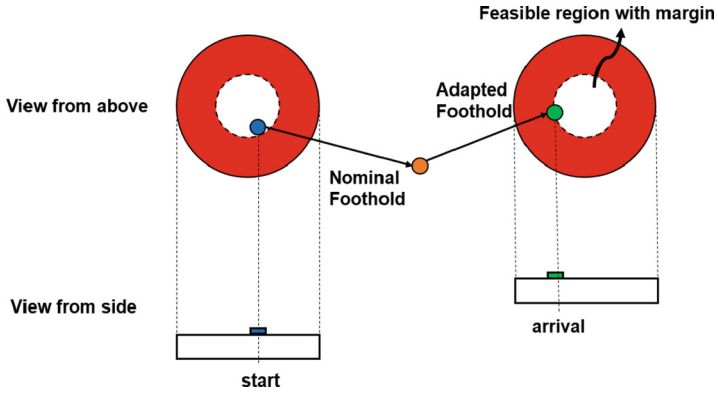

Fig. 3. Illustration of the foothold adaptation process, the nominal foothold position is adjusted to the nearest feasible region.

$$\mathbf{p}_{foot} = \mathbf{p}_{hip} + \frac{T_s}{2}\mathbf{v}_{base}, \tag{1}$$

where T_s is the stance phase time, and $\mathbf{v}_{base}$ is the base velocity. However, it cannot be guaranteed that there is feasible ground to set foot on at that location. Therefore, the calculated foothold position is moved to the nearest feasible region if infeasible. Fig. 3 visualizes the process of adjustment from the nominal foothold computed from Raibert's heuristic to the adapted foothold. At this time, the feasible region is determined by placing a 5 cm safety margin parallel to the boundary of the land that can be stepped on.

The foothold position is not pre-determined in the path planning stage but is determined reactively according to the current hip's position and the base's speed. If the torso reference and the foothold reference are derived together in the path-planning stage, they are correlated with each other. Hence, the foothold position is feasible only when the base tracks the torso reference well. This causes instability when the foot follows the foothold reference, but the robot's base is out of torso reference due to disturbance or uncertainty. The reactive foothold decision prevents the feet from moving ahead of the body and causing instability. The calculated next foothold position is reflected in the forward pass process of the MPC.

4.2 Swing Trajectory

The foothold position determined in Sect. 4.1 corresponds to the arrival point of the swing trajectory. After selecting the start and arrival of the swing trajectory, a curve connecting them must be designed. Fig. 4(a) and 4(b) show how the swing trajectory is generated in the upstairs and downstairs situations, respectively. The layover of the swing trajectory should be located near arrival when going down the step and near the start when going up the step so that the legs do not collide with the terrain. We define the layover weight alpha as a modified

sigmoid function for the height difference between arrival and start. From the layover weight, the position of the layover is determined as follows.

$$x_{layover} = \alpha x_{start} + (1-\alpha)x_{arrival} \tag{2}$$
$$y_{layover} = \alpha y_{start} + (1-\alpha)y_{arrival} \tag{3}$$
$$z_{layover} = (1-\alpha)z_{start} + \alpha z_{arrival}, \tag{4}$$

where x, y, and z are components of a 3-D position in the global frame. Then, the 11th-order bezier curve defined by start, layover, and arrival is set as the swing trajectory reference. This swing trajectory does not contain hard constraints to ensure collision avoidance, but it works well in general non-flat terrain.

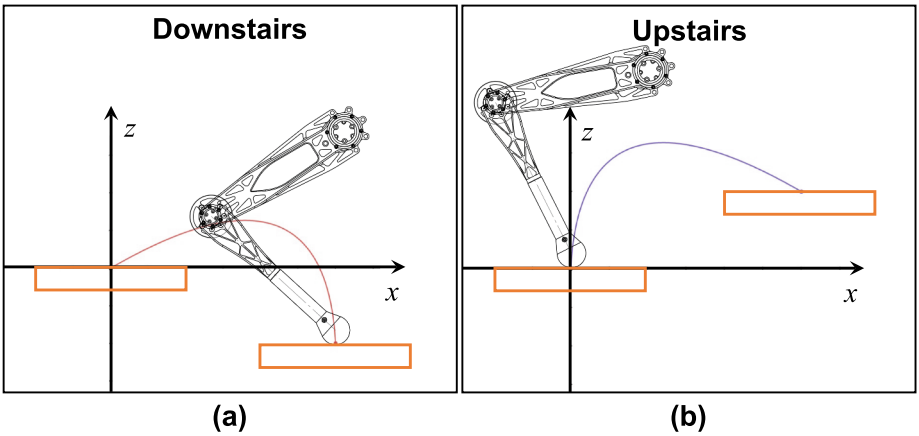

Fig. 4. Illustration of the adjusting swing leg trajectory to ensure collision-free with terrain.

5 Results

The evaluation was conducted on maps under various conditions to study the proposed path planner and reference generator's performance. The conditions of the MPC parameters remain constant. First, the results of path planning based on RRT* are shown in Sect. 5.1. Second, Sect. 5.2 explains the results of a task in which a robot traverses a challenging terrain based on the path planner and reference generator. Finally, problems arising from traversing tasks are addressed in Sect. 5.3.

5.1 Path Planning

Figure 5 provides path planning results of various maps. For all three cases, the steer length, the distance to create a new node from an existing node in the path search process, is the same as 0.1 m. From the path planning, a feasible path for the robot to pass from the initial point to the goal point was derived. In addition to avoiding terrain that cannot be crossed due to a lack of stepping stones, feasible and infeasible paths are distinguished kinematically. It does not cross over large gaps or navigate points where edges meet. For example, in Fig. 5(a), passing through the narrow route between holes seems to be the shortest path. However, the path planner considers this to be kinematically infeasible and instead creates a curved way that avoids the holes. As we checked on the map in several cases, the created paths were all smooth enough for the robot to track. In a map with less sparsity, a path close to a straight line is created instead of an unnecessary curved path. On the other hand, in a map with large sparsity, a curved path is made to avoid holes to satisfy kinematic feasibility. About 2,000 nodes are required to complete path planning in a map environment with a size of 4 m in width and 8 m in length.

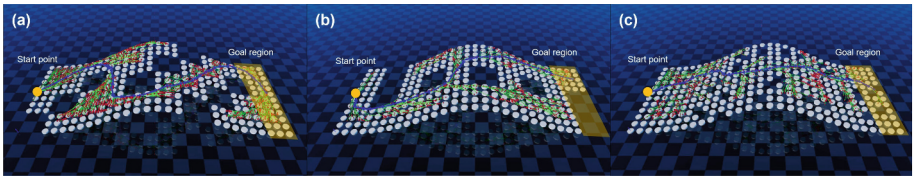

Fig. 5. Results of the path planning (RRT*) in diverse maps. The blue line represents the path that has been identified using the path planning algorithm. The green line indicates the edges connecting the nodes in the path, while the red line represents the rewired edges. As shown in the figure, the planner successfully found the kinematic feasible path from the initial point to the goal region, avoiding the empty region.

5.2 Traversing Challenging Terrain

In the traversing challenging terrain task, a steer length of 0.4 m was applied to path planning. Fig. 6 shows that the robot walks along a given node through foothold adaptation in a situation where the sparsity of the stepping stone exists. The terrain is generated by distributing stepping stones based on a Gaussian function and creating gaps in certain areas, making it challenging to traverse the terrain with a simple straight path. If only the simple Raibert's heuristic were used, the robot easily fall as it could step into the air (not a stepping stone) and lose its balance. However, the robot can pass through sparse terrain because the arrival point of the swing trajectory is always set to the feasible ground through foothold adaptation. Additionally, the foothold adaptation is based on

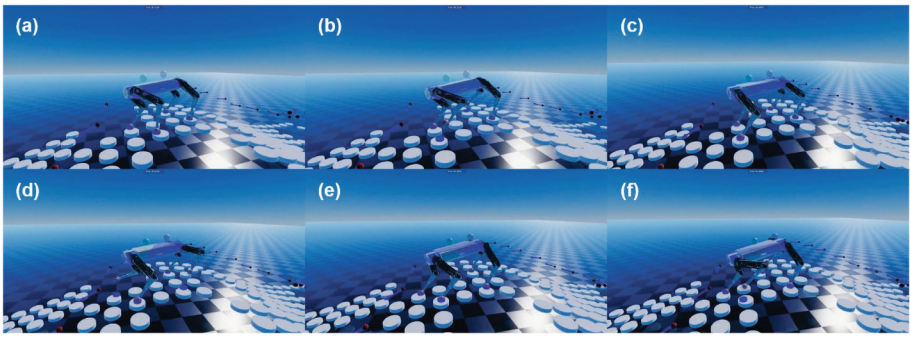

Fig. 6. Simulation results of the whole framework. The robot follows the generated nodes (represented by red dots within its height range). The snapshots shown in panels (a) to (f) illustrate the robot's progression over time. The robot adapts its next foothold position (indicated by blue disks) to the stepping stone, ensuring proper ground reaction forces are generated by MPC in the designated foothold positions.

Raibert's heuristic, the robot doesn't step too far or too close to the robot's hip position unnecessarily. As the robot follows the body reference trajectory marked by the red dot in Fig. 6(b), it adapts the Front Left (FL) foothold by moving it inward due to the absence of a valid foothold in the outward direction. Additionally, in Fig. 6(f), the robot restricts the forward movement of the FL leg to ensure dynamic stability when it goes descending direction while allowing the Rear Right (RR) foothold to move forward.

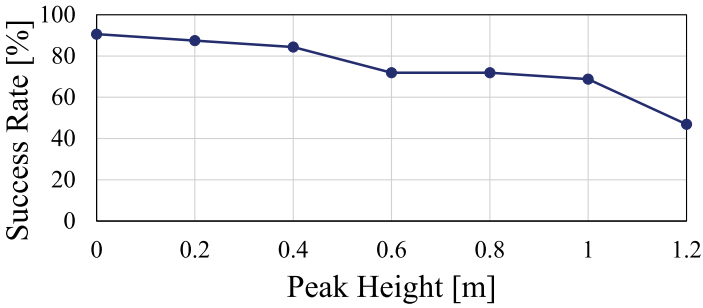

Fig. 7. Success rate of traversal based on terrain difficulty.

Figure 7 shows the robot's success rate of traversal based on terrain difficulty, specifically the peak height of Gaussian function-based terrain. As the peak height increases, the terrain becomes steeper. The robot's starting position and gap locations were randomly assigned to measure an average success rate. For successful traversal, several conditions must be satisfied together: a feasible path needs to be identified, the MPC accurately tracks this path to prevent the robot

from falling, and foothold adaptation also functions effectively. Considering these factors, the overall results demonstrate a reasonable success rate. Despite these achievements, the success rate is highly dependent on terrain conditions such as slope and sparsity. The reasons for this are discussed in the next subsection.

5.3 Limitations

The fundamental challenge during traversing tasks is that the path planner primarily focuses on achieving kinematic feasibility while neglecting dynamic feasibility considerations. Even though the robot tries to capture its dynamic feasibility through foothold adaptation, if the planned trajectory is too hard to follow dynamically (e.g. rapid direction change), the robot cannot stabilize for this commanded path. In other words, the reference generator creates a reference trajectory simply by linear interpolating x, y, and yaw between nodes, dynamic stability during walking is not guaranteed with only the proposed path planner and reference generator. Although our framework addresses this issue to some extent through foothold adaptation, it may not be sufficient for tackling more challenging environments, such as steeper stairs. As a result, although path planning generally derives a smooth path, in our experiments, In some cases, the significant difference in yaw commands between nodes can lead to instability issues when the robot tries to track them.

Another limitation of this study is that many terrains cannot be overcome because a given node is tracked with a constant gait pattern and speed. For example, terrain that cannot be reached by walking gait but can be passed by jumping is hard to overcome with the current control structure.

6 Conclusion

This paper proposes a hierarchical framework for enhancing the locomotion capabilities of legged robots in challenging terrains. It combines CoM path planning using the RRT* algorithm with local foothold adaptation. The RRT* algorithm creates a path from the initial point to the goal region by considering kinematic feasibility, where the potential foothold positions are valid with corresponding body configurations. While tracking the path, the robot made it step on the stepping stone through foothold adaptation and prevented collision by applying layover weight to the swing trajectory. The proposed method demonstrated successful route creation and traversal in various maps with sparse stepping stones. Future work will focus on expanding this approach by incorporating dynamic feasibility into the path planning process, enabling the robot to overcome even more arbitrary and challenging terrains.

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